

MICHAEL WHITEMAN

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Education

University of Waterloo

Undergraduate in Computer Engineering

Sep. 2023 – Present

Waterloo, Ontario

Technical Skills

Languages: C, C++, C#, Python, JavaScript, HTML/CSS, ARM, RISC-V ASM, Verilog, VHDL

Technologies: Linux, Windows, OpenGL, Git/GitHub, STM32CubeIDE, Arduino, Quartus Prime, Raspberry Pi 4, RISC-V, Valgrind, React, Vue, MySQL, SQLite3, JWT, AutoCAD, VSCode, Visual Studio

Hardware Skills: 3D Printing, PCB Building, Soldering, Multimeter, Oscilloscope, DMM, Function Generator

Experience

Software Developer | *React, JS, C++, MySQL, SQLite3, JWT, RPI4*

Jan 2026 – Apr 2026

Pinkbyte

Concord, Ontario

- Built a **fullstack web application** using React, JavaScript, C++, **MySQL**, and **SQLite3**, with a live database interface enabling admins to connect, query, and modify records through the web UI.
- Implemented a **multithreaded alert system** in **C++** to handle real-time event notifications across concurrent modules, integrated with a 3rd party service to send alerts to company emails.
- Developed** secure user authentication with **JWT**-based login and session management for a fullstack internal web platform.
- Prototyped wireless **serial communication** via Raspberry Pi 4, parsing input commands and **translating** them into **opcodes** dispatched to a HERO robot over serial interface.

Coding Instructor | *C#, C++, JavaScript, Lua, Unity, Roblox Studio*

Sept 2024 – Dec 2024

Code Ninjas

Richmond Hill, Ontario

- Taught students C++ ranging from basics to high-efficiency algorithms, preparing custom lessons and coding problems to help prepare them for the **Waterloo CCC** contest.
- Mentored **30+** students ranging from ages 6–13 through C++, C#, Lua, and JavaScript, and game development using **Roblox Studio** and **Unity**.
- Created material for other instructors and ran multiple workshops on **3D modelling**, 3D printing, Roblox game creation, and modding in Minecraft.

Projects

Real-Time Kernel (RTX) | *C, ARM Cortex-M4, RTOS, STM32CubeIDE, Gitlab*

Apr 2026 – Present

- Built a preemptive **real-time kernel** from scratch in **C** on **ARM Cortex-M4**, implementing task creation, PCB/TCB management, and context switching via custom ARM assembly routines.
- Designed** a fixed-size memory pool allocator to prevent heap fragmentation in time-critical kernel code paths.
- Implemented priority-based and **EDF (Earliest Deadline First)** scheduling to guarantee deterministic real-time task execution under strict timing constraints.

Embedded FPGA Audio Player | *C, Altera MAX10 FPGA, Quartus Prime, NIOS II*

Oct 2025 – Dec 2025

- Developed modular firmware (loopback, UART CLI, button interface, WAV player) in C using Altera Eclipse SBT.
- Employed **SignalTap II** for signal viewing and logic analyzer for timing and synchronization verification.
- Multiplexed** dual 7-segment displays to show converted values, **improving readability on constrained hardware**.

LTE RC Car | *C++, Linux, AutoDesk Fusion, Github, Raspberry Pi 4, Soldering*

June 2025 – Aug 2025

- Developed a **C++** client-server network architecture to **stream real-time inputs** and telemetry with low latency.
- Designed and assembled an LTE-controlled RC car using a Raspberry Pi 4 and SIM7600 module to enable **remote operation over effectively unlimited range**.
- Implemented **PWM-based motor and servo control** using custom software and hardware PWM signals.
- Applied **soldering and wiring techniques** to connect and to guarantee functionality of all electronic components.

Go Luxury Cars Management Site | *C++, JavaScript, Vue, Postman API*

Sept 2024 – Dec 2024

- Developed and debugged a **management website** for Go Luxury Cars to organize client management, order tracking, and inventory using C++ for backend and **Vue** for frontend.
- Connected **frontend and backend** aspects using **Postman API** and Vue to ensure visual data is displayed properly and synchronized in real time.
- Researched runtime optimization strategies and organized coding files to support **future co-op students**.